

Hamiltonian Cycle(H) main()
 { Hc(2);
 }

Initially $X[1] = 1$
 $X[2:n] = 0$

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→ Hamiltonian_Cycle(k)

```
{
do
{
  NextValue(k);
  if (X[k] == 0)
    return;
  else
  {
```

```
    if (k == n && G[X[n]][X[1]] == 1)
    {
```

```
      Print X[1:n];
    }
```

```
    else if (k < n)
      Hamiltonian_Cycle(k+1);
  }
```

```
} while (true);
```

```
}
```

→ NextValue(k)

```
{ do
{
  X[k] = (X[k] + 1) % (n + 1);
  if (X[k] == 0)
    return;
```

```
else
{ if (G[X[k-1]][X[k]] == 1)
```

```
{ for (j = 1; j <= k-1; j++)
```

```
{ if (X[j] == X[k])
{ break;
}
```

```
}
```

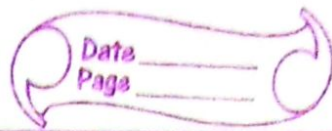
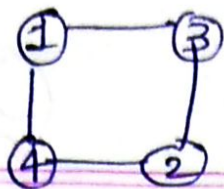
```
if (j == k)
```

```
return;
```

```
} } while (true);
```

- Here $X[k]$ denotes k^{th} vertex in the Hamiltonian cycle
- Assume start of cycle is always 1
 ∴ We called from 2

Tracing:



Initially $X = [1 \ 0 \ 0 \ 0]$

$H(2)$

— $NV(2)$

- $X[2] = 1$

$G[X[1]][X[2]]$

$= G[1][1] \neq 1$

$\therefore$ Increment $X[2]$

- $X[2] = 2$

$G[X[1]][X[2]]$

$= G[1][2] = 0 \neq 1$

$\therefore$ Increment $X[2]$

- $X[2] = 3$

$G[X[1]][X[2]]$

$= G[1][3] = 1$

$\therefore X[2] = 3$ which is different from $X[1]$

$\therefore j = k$ will be true and it returns to $H(2)$

$X[2] = 3$

$X = [1 \ 3 \ 0 \ 0]$

$2 < 3$ (i.e. $k < n$)

$\therefore$ It will call $H(3)$

$H(3)$

— $NV(3)$

- $X[3] = 1$

$G[X[2]][X[3]]$

$= G[3][1] = 1$

$\therefore$ check if its distinct

$j = 1$ $X[1] = X[3]$ i.e. $1 = 1 \Rightarrow$ break for loop

- $X[3] = 2$

$G[X[2]][X[3]] = G[3][2] = 1$ $\therefore$ Check $j = 1, 2$ and

it will not break $\therefore X = [1 \ 3 \ 2 \ 0]$ & return to

$3 < 4 \Rightarrow$ call $H(4)$

$H(3)$

H(4)

— NV(4)

• $X[4] = 1$

$G[X[3]][X[4]]$
 $= G[2][1] \neq 1$

• $X[4] = 2$

$G[X[3]][X[4]]$
 $G[2][2] \neq 1$

• $X[4] = 3$

$G[X[3]][X[4]]$
 $G[2][3] = 1$

$j=1 \quad X[1] \neq X[4]$

$j=2 \quad X[2] \neq X[4] \text{ (i.e. } 3 \neq 3)$

∴ break loop & continue to while loop.

• $X[4] = 4$

$G[X[3]][X[4]]$
 $G[2][4] = 1$

$j=1 \quad X[1] \neq X[4] \quad 1 \neq 4$

$j=2 \quad X[2] \neq X[4] \quad 3 \neq 4$

$j=3 \quad X[3] \neq X[4] \quad 2 \neq 4$

∴ $j=4$ → return to H(4)

∴ $X = [1 \ 3 \ 2 \ 4]$

— Here $k=n$ & $G[4][1] = 1$

∴ print $[1 \ 3 \ 2 \ 4]$ i.e. solution vector.

— Again call NV(4)

$X[4] = (4+1) \% 5 = 0 \Rightarrow$ return to H(4)

↳ In $H(4)$, $X[4] == 0 \Rightarrow$ return to caller

i.e. $H(3)$'s Activation

★ In $H(3)$ [Again do while loop will run]

NV(3)

$$\begin{aligned} \bullet X[3] &= (X[3] + 1) / 5 \\ &= (2 + 1) / 5 \\ &= 3 \end{aligned}$$

$$K[X[2]][X[3]]$$

$$K[3][3] \neq 1 \Rightarrow \text{Increment } X[3]$$

$$\begin{aligned} \bullet X[3] &= (3 + 1) / 5 \\ &= 4 \end{aligned}$$

$$K[X[2]][X[3]]$$

$$= K[3][4] \neq 1 \Rightarrow \text{Inc. } X[3]$$

$$\begin{aligned} \bullet X[3] &= (4 + 1) / 5 \\ &= 0 \end{aligned}$$

Since $X[3] == 0 \Rightarrow$ return to $H(3)$.

In $H(3)$,

Since $X[3] == 0 \Rightarrow$ return to $H(2)$

$$X = [1 \ 3 \ 0 \ 0]$$

★ In $H(2)$.

NV(2)

$$\bullet X[2] = (3 + 1) / 5 = 4$$

$$K[X[1]][X[2]]$$

$$= K[1][4] = 1$$

$$\therefore j = 1 \quad X[1] \neq X[2]$$

$$1 \neq 4 \Rightarrow X = [1 \ 4 \ 0 \ 0]$$

$j = 2 \Rightarrow$ return to $H(2)$

$2 < 4 \therefore$ call $H(3)$

↳ $H(3)$

$$HC(3) \quad X = [1 \ 4 \ 0 \ 0]$$

$$NV(3)$$

$$\bullet X[3] = (0+1)/.5 = 1$$

$$\bullet G[X[2]][X[3]] = G[4][1] = 1$$

$$\therefore j = 1 \quad X[1] = X[3] \quad 1 = 1 \Rightarrow \text{break;}$$

$$\bullet X[3] = (1+1)/.5 = 2$$

$$G[X[2]][X[3]]$$

$$G[4][2] = 1$$

$$\therefore j = 1 \quad X[1] \neq X[3] \quad 1 \neq 2$$

$$j = 2 \quad X[2] \neq X[3] \quad 4 \neq 2$$

$$j = 3 \Rightarrow \text{return to } \therefore X = [1 \ 4 \ 2 \ 0]$$

$$HC(3)$$

$$3 < 4 \therefore \text{call } HC(4)$$

$$HC(4)$$

$$NV(4)$$

$$\bullet X[4] = (0+1)/.5 = 1$$

$$G[X[3]][X[4]]$$

$$= G[2][1] \neq 1$$

$$\bullet X[4] = 2/.5 = 2$$

$$G[X[3]][X[4]]$$

$$= G[2][2] \neq 1$$

$$\bullet X[4] = 3/.5 = 3$$

$$G[X[3]][X[4]] \quad X = [1 \ 4 \ 2 \ 3]$$

$$= G[2][3] = 1$$

$\therefore$ Check for every j

$$j=1 \quad X[1] \neq X[4] \quad 1 \neq 3$$

$$j=2 \quad X[2] \neq X[4] \quad 4 \neq 3$$

$$j=3 \quad X[3] \neq X[4] \quad 2 \neq 3$$

$\therefore j=4 \Rightarrow$ return to $H(4)$

$\rightarrow H(4)$

Now $k==n$ && $G[X[n]][X[1]]==1$

$4==4$ && $G[X[4]][1]==1$

$4==4$ && $G[3][1]==1$

$\downarrow$

True

$\downarrow$

True

$\therefore$ print another solution [1 4 2 3]

continue in do-while loop

$\rightarrow NV(4)$

$$\bullet X[4] = (3H)/.5$$

$$= 4$$

$$j=1 \quad X[1] \neq X[4] \quad 1 \neq 4$$

$$j=2 \quad X[2] == X[4] \quad 4 == 4 \Rightarrow \text{break}$$

$\therefore$ Increment $X[4]$ in do-while

$$\bullet X[4] = (4H)/.5$$

$$= 0$$

$\therefore$ return to $H(4)$

$$\rightarrow X[4] == 0$$

$\therefore$ return to $H(3)$

X [1 4 2 0]

$\rightarrow$ In $H(3)$,

$NV(3)$

$$\bullet X[3] = (2H)/.5 = 3 \quad [1 \ 4 \ 3 \ 0]$$

$$G[X[2]][X[3]] = G[4][3] \neq 1$$

$$\bullet X[3] = (3H)/.5 = 4 \quad [1 \ 4 \ 4 \ 0]$$

$$G[X[2]][X[3]] \\ = G[4][4] \neq 1$$

$$\bullet X[3] = (4+1)/.5 \\ = 0$$

$$[1 \ 4 \ 0 \ 0]$$

return to H(3)

→ H(3) returns to H(2)

In. H(2), ^{In} NV(2)

$$X[2] = (4+1)/.5 \\ = 0$$

$$[1 \ 0 \ 0 \ 0]$$

∴ Return to H(2)

From H(2) return to ~~H(1)~~ main()

~~In H(1), In NV(1)~~

Therefore, It has printed two solⁿs

$$[1 \ 3 \ 2 \ 4]$$

$$[1 \ 4 \ 2 \ 3]$$